



POWER BI MAJOR CAPSTONE PROJECT
END TO END
POWER BI PROJECT GUIDE

PROJECT TITLE	SCHOOL INFRASTRUCTURE C ENROLLMENT ANALYTICS DASHBOARD
PROJECT CATEGORY	Education Analytics School Analytics Infrastructure Analytics
INSTITUTE NAME	Infinity Code Nexus
TRAINER / GUIDE	Anamania Anand Kumar
CREATED BY	Y Sai Somnath Guptha
TOOLS USED	Power BI Desktop Power Query, DAX C Power BI Service
SUBMISSION DATE	[10/06/2026]

Prepared as part of the Power BI Masterclass Program to demonstrate skills in Data Analysis, Data Modeling, DAX Development, Dashboard Design, UI/UX, Storytelling, and Business Intelligence Reporting.

PROJECT OVERVIEW

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PROJECT OBJECTIVE

Develop a comprehensive Power BI dashboard to analyze sales performance, profit trends, customer behavior, and product performance across different regions and market segments. The dashboard provides insights into revenue generation, profitability, customer contributions, and geographical sales distribution. It helps businesses make data-driven decisions, optimize strategies, and improve overall operational performance.

Participants are expected to:

- ✓ Analyze sales performance across regions and market segments
- ✓ Evaluate customer purchasing behavior and revenue contribution
- ✓ Assess product performance, profitability, and category trends
- ✓ Compare sales, profit, and customer distribution across locations
- ✓ Generate meaningful business insights through interactive visualizations
- ✓ Present recommendations for improving sales growth and profitability

PROJECT GOAL

Convert raw sales data into actionable insights that help understand business performance, customer behavior, product profitability, and regional sales distribution. The dashboard enables data-driven decision-making by identifying key trends, opportunities, and areas for business improvement.

BUSINESS PROBLEM

Business managers and decision-makers need visibility into sales performance, customer behavior, product profitability, and regional business operations.

The objective is to identify:

- Which regions generate the highest sales and profit?
- Which products contribute the most revenue and profitability?
- How are customers distributed across different segments and regions?
- Whether discounts are impacting profitability?

- How sales and profit trends vary over time?
- Which customer segments and product categories drive business growth?
- How business performance varies across different locations and markets?

BUSINESS QUESTIONS

Participants must answer:

1. How much total sales revenue has the business generated?
2. Which regions contribute the highest sales and profit?
3. Which product categories and sub-categories dominate sales performance?
4. What is the total number of customers and orders?
5. Which customers generate the highest revenue?
6. Which products deliver the highest profit and which products incur losses?
7. How do sales and profit trends vary across months and years?
8. What is the impact of discounts on profitability?
9. Which states and regions require business improvement?
10. Which customer segments contribute most to overall business growth?

DATASET INFORMATION

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DATASET NAME : Superstore Sales and Profit

DATASET DETAILS : Approximate Records 9,994+

Columns : 19

EXPECTED ANALYSIS AREAS

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Executive Overview

Sales Analytics

Profit Analytics

Customer Analytics

Product Analytics

POWER QUERY TRANSFORMATIONS

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DATA CLEANING ACTIVITIES

- ✓ Validated sales transaction records
- ✓ Handled missing and blank values
- ✓ Corrected data types for Date, Sales, Quantity, Discount, and Profit columns
- ✓ Standardized Region, State, and Category values
- ✓ Verified Sales, Profit, and Quantity records
- ✓ Removed duplicate records and inconsistencies
- ✓ Validated customer and product information

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DATA PREPARATION ACTIVITIES

- ✓ Renamed columns for better readability
- ✓ Replaced blank and null values where applicable
- ✓ Corrected date and numerical formats
- ✓ Removed unnecessary columns from supporting sheets
- ✓ Validated sales and profit calculations
- ✓ Created relationships between fact and dimension tables
- ✓ Built Date Table for time intelligence analysis
- ✓ Prepared data for dashboard visualization and analysis

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CUSTOM COLUMNS CREATED

- ✓ Profit Margin Percentage
- ✓ Customer Value Category
- ✓ Sales Growth Percentage
- ✓ Product Sales Rank
- ✓ Customer Revenue Rank
- ✓ Regional Performance Category
- ✓ Order Value Category
- ✓ Monthly Sales Classification
- ✓ Product Profitability Category

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SCREENSHOT

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State	Postal Code	Region	Product ID
43	CA-2016-101343	17 July 2016	22 July 2016	Standard Class	RA-19805	Ruben Ausman	Corporate	United States	Los Angeles	California	90049	West	OFF-ST-10003
514	CA-2017-163405	21 December 2017	25 December 2017	Standard Class	BN-15151	Bradley Nguyen	Consumer	United States	Los Angeles	California	90049	West	OFF-AR-10001
515	CA-2017-163405	21 December 2017	25 December 2017	Standard Class	BN-15151	Bradley Nguyen	Consumer	United States	Los Angeles	California	90049	West	OFF-AR-10001
1626	CA-2015-157084	19 December 2015	24 December 2015	Standard Class	JG-15160	James Gallang	Consumer	United States	Los Angeles	California	90049	West	TECH-AC-100
2060	CA-2014-106439	31 October 2014	04 November 2014	Standard Class	GG-14650	Greg Guthrie	Corporate	United States	Los Angeles	California	90049	West	OFF-FA-10002
2067	CA-2014-106439	31 October 2014	04 November 2014	Standard Class	GG-14650	Greg Guthrie	Corporate	United States	Los Angeles	California	90049	West	OFF-ST-10003
2065	CA-2014-106439	31 October 2014	04 November 2014	Standard Class	GG-14650	Greg Guthrie	Corporate	United States	Los Angeles	California	90049	West	OFF-ST-10000
2067	CA-2014-106439	31 October 2014	04 November 2014	Standard Class	GG-14650	Greg Guthrie	Corporate	United States	Los Angeles	California	90049	West	TECH-AC-100
2068	CA-2014-106439	31 October 2014	04 November 2014	Standard Class	GG-14650	Greg Guthrie	Corporate	United States	Los Angeles	California	90049	West	OFF-AR-10000
2791	CA-2016-118913	25 June 2016	29 June 2016	Standard Class	AS-10240	Alan Shonely	Consumer	United States	Los Angeles	California	90049	West	OFF-AP-10000
2709	CA-2015-121197	30 January 2015	06 February 2015	Standard Class	CC-12145	Charles Crestani	Consumer	United States	Los Angeles	California	90049	West	FUR-FU-10001
2944	CA-2017-126242	19 November 2017	24 November 2017	Standard Class	MC-18100	Mick Crebagga	Consumer	United States	Los Angeles	California	90049	West	OFF-ST-10000
2945	CA-2017-126242	19 November 2017	24 November 2017	Standard Class	MC-18100	Mick Crebagga	Consumer	United States	Los Angeles	California	90049	West	FUR-FU-10000
3534	CA-2014-110849	18 April 2014	23 April 2014	Standard Class	IL-15835	John Lee	Consumer	United States	Los Angeles	California	90049	West	OFF-AR-10001
3535	CA-2014-110849	18 April 2014	23 April 2014	Standard Class	IL-15835	John Lee	Consumer	United States	Los Angeles	California	90049	West	OFF-AR-10000
3536	CA-2014-110849	18 April 2014	23 April 2014	Standard Class	IL-15835	John Lee	Consumer	United States	Los Angeles	California	90049	West	OFF-FA-10000
3759	CA-2015-167745	18 September 2015	23 September 2015	Standard Class	GB-14530	George Bell	Corporate	United States	Los Angeles	California	90049	West	OFF-FA-10001
3760	CA-2015-167745	18 September 2015	23 September 2015	Standard Class	GB-14530	George Bell	Corporate	United States	Los Angeles	California	90049	West	OFF-SU-1000+
3761	CA-2015-167745	18 September 2015	23 September 2015	Standard Class	GB-14530	George Bell	Corporate	United States	Los Angeles	California	90049	West	FUR-FU-10001
4330	CA-2015-125234	27 November 2015	01 December 2015	Standard Class	SN-20710	Steve Nguyen	Home Office	United States	Los Angeles	California	90049	West	TECH-AC-100
4339	CA-2015-125234	27 November 2015	01 December 2015	Standard Class	SN-20710	Steve Nguyen	Home Office	United States	Los Angeles	California	90049	West	OFF-FA-10000
4481	CA-2016-109365	03 November 2016	08 November 2016	Standard Class	XP-21865	Xylona Preis	Consumer	United States	Los Angeles	California	90049	West	OFF-FA-10001
4482	CA-2016-109365	03 November 2016	08 November 2016	Standard Class	XP-21865	Xylona Preis	Consumer	United States	Los Angeles	California	90049	West	OFF-AP-10000
4483	CA-2016-109365	03 November 2016	08 November 2016	Standard Class	XP-21865	Xylona Preis	Consumer	United States	Los Angeles	California	90049	West	OFF-FA-10001
4484	CA-2016-109365	03 November 2016	08 November 2016	Standard Class	XP-21865	Xylona Preis	Consumer	United States	Los Angeles	California	90049	West	OFF-PA-10000
4485	CA-2016-109365	03 November 2016	08 November 2016	Standard Class	XP-21865	Xylona Preis	Consumer	United States	Los Angeles	California	90049	West	OFF-PA-10000
4487	CA-2016-109365	03 November 2016	08 November 2016	Standard Class	XP-21865	Xylona Preis	Consumer	United States	Los Angeles	California	90049	West	OFF-PA-10000

DATA MODELING

MODEL DESIGN

Purpose:

The data model was designed to support interactive analysis of sales performance, customer behavior, product profitability, and regional business distribution. The model enables efficient reporting, filtering, trend analysis, and business intelligence dashboard development.

MAIN TABLE

Superstore Sales Dataset

KEY FIELDS

- Order ID
- Order Date
- Ship Date
- Customer ID
- Customer Name
- Segment
- Region
- State
- City
- Product ID
- Category

Sub-Category
Product Name
Sales
Quantity
Discount
Profit

RELATIONSHIPS

Single Table Data Model

(No additional relationships were required because the dataset was maintained in a structured and consolidated format containing customer, product, sales, and geographical information.)

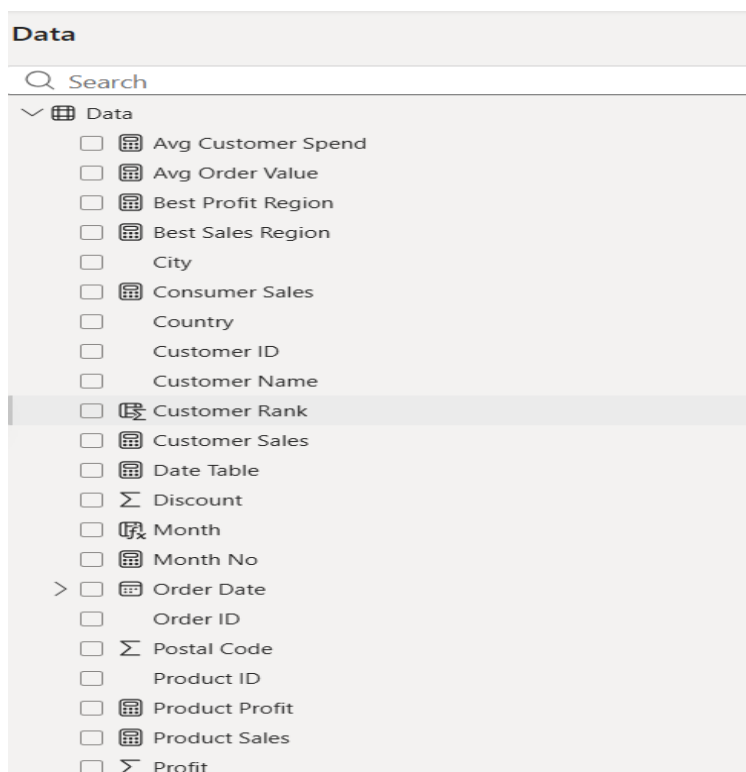
FILTER DIRECTION

Single Direction

MODEL BENEFITS

- ✓ Improved Report Performance
- ✓ Easy DAX Development
- ✓ Simplified Dashboard Design
- ✓ Efficient Data Analysis
- ✓ Interactive Filtering and Drill Analysis
- ✓ Better User Experience

SCREENSHOT



DAX MEASURES USED

TOTAL ORDERS

Total Orders =

DISTINCTCOUNT(Data[Order ID])

Purpose:

Calculate the total number of unique orders placed.

TOTAL SALES

Total Sales =

SUM(Data[Sales])

Purpose:

Calculate the total sales revenue generated across all orders.

TOTAL PROFIT

Total Profit =

SUM(Data[Profit])

Purpose:

Calculate the total profit earned from all sales transactions.

TOTAL CUSTOMERS

Total Customers =

DISTINCTCOUNT(Data[Customer ID])

Purpose:

Calculate the total number of unique customers.

TOTAL PRODUCTS

Total Products =

DISTINCTCOUNT(Data[Product ID])

Purpose:

Calculate the total number of unique products sold.

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AVERAGE ORDER VALUE

Average Order Value =
DIVIDE([Total Sales],[Total Orders],0)

Purpose:

Calculate the average revenue generated per order.

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PROFIT MARGIN %

Profit Margin % =
DIVIDE([Total Profit],[Total Sales],0)

Purpose:

Calculate the percentage of profit earned from total sales.

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TOP SALES REGION

Top Sales Region

Purpose:

Identify the region generating the highest sales revenue.

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TOP SALES REGION

Total Pre Primary Rooms =
SUM('Basic Details of Schools_Sample_Data'[Pre Primary Rooms])

Purpose:

Calculate total pre-primary classrooms.

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VISUALIZATIONS USED

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KPI CARDS

Purpose:

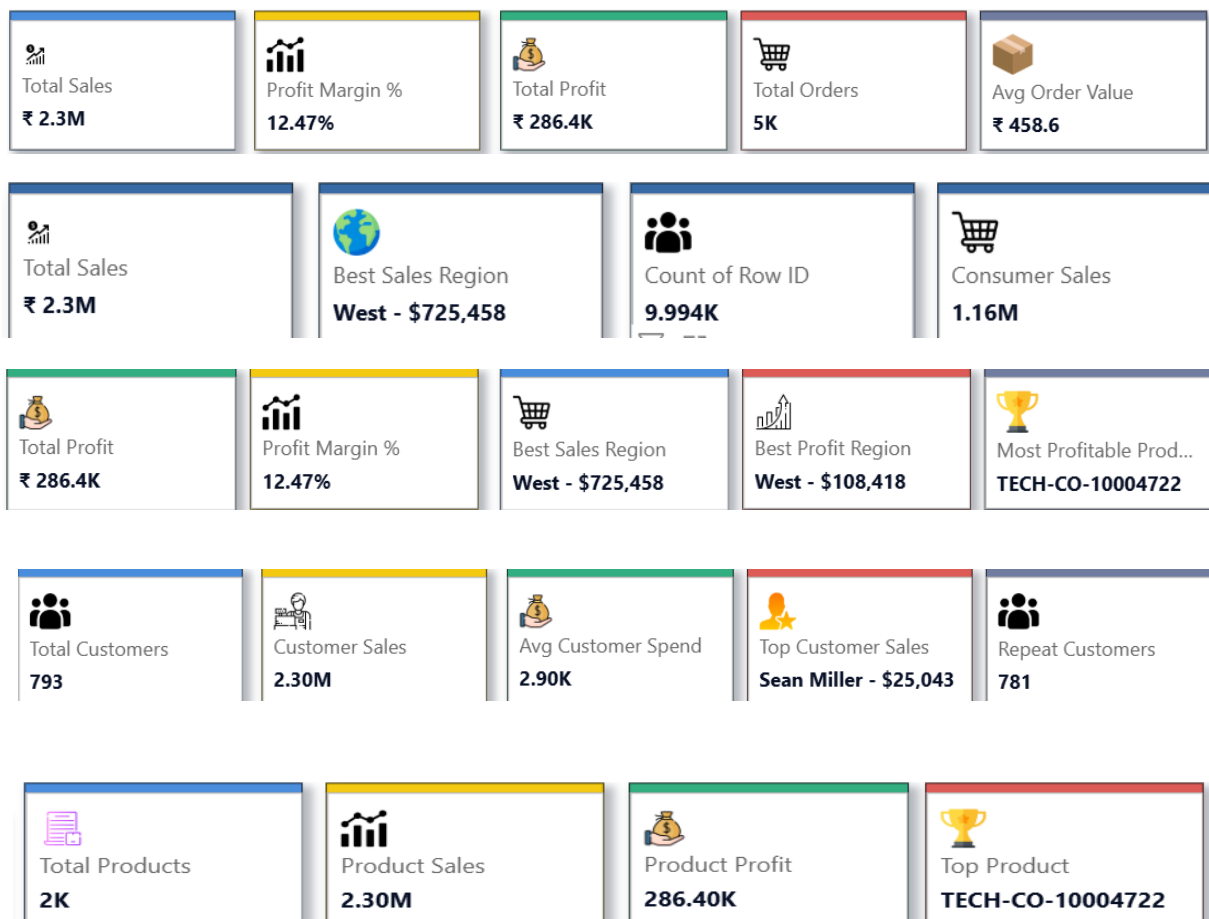
KPI Cards are used to display key business metrics and provide a quick overview of sales performance, profitability, customer activity, product performance, and regional business growth.

KPIs Included:

- Total Sales
- Total Profit

- Total Orders
- Total Customers
- Total Products
- Average Order Value
- Top Sales Region
- Top Profit Region
- Top Customer
- Top Product
- Average Customer Spend
- Repeat Customers
- Highest Sales Value
- Highest Profit Value

SCREENSHOT



CLUSTERED BAR CHART

Purpose:

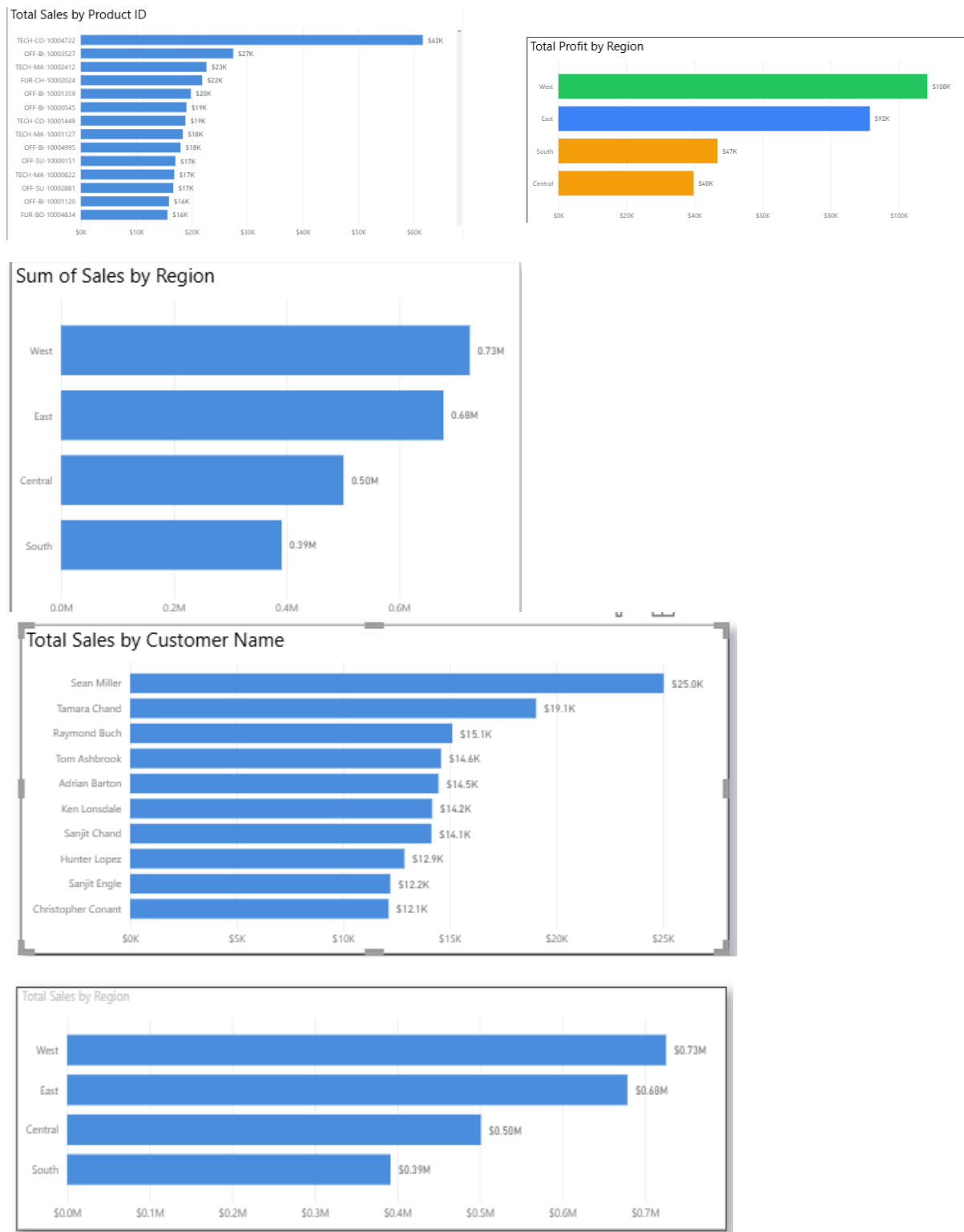
Clustered Bar Charts are used to compare sales, profit, customer, and product performance across regions, categories, states, and customer segments.

The charts were used to analyze:

- Sales by Region
- Profit by Region
- Top 10 Customers by Revenue
- Top 10 Products by Sales
- Top 10 Products by Profit
- Sales by Category
- Profit by Category

These visualizations help identify top-performing products, customers, categories, and regions while highlighting areas that require business improvement.

SCREENSHOT



CLUSTERED COLUMN CHART

Purpose:

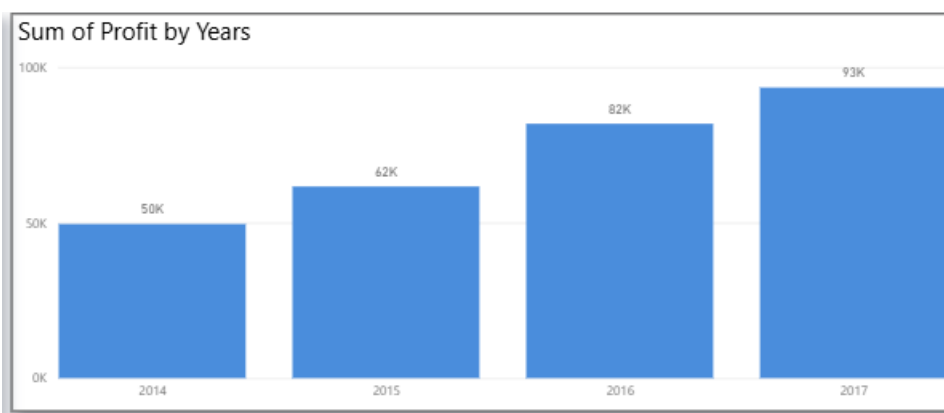
Clustered Column Charts are used to compare sales, profit, customer, and product performance across different categories, regions, and customer segments.

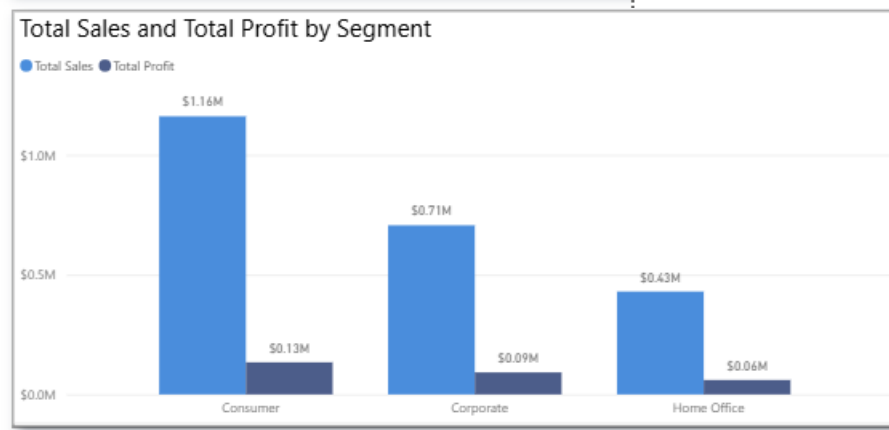
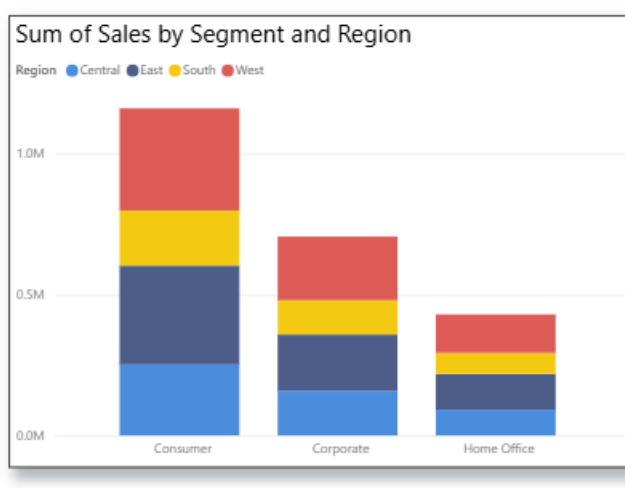
The charts were used to analyze:

- Product Sales by Category
- Quantity Sold by Category
- Sales by Region
- Profit by Region
- State-wise Sales Performance
- Monthly Sales Comparison

These visualizations help compare business performance across categories, products, customers, and regions while identifying growth opportunities and improvement areas.

SCREENSHOT





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DONUT CHART

Purpose:

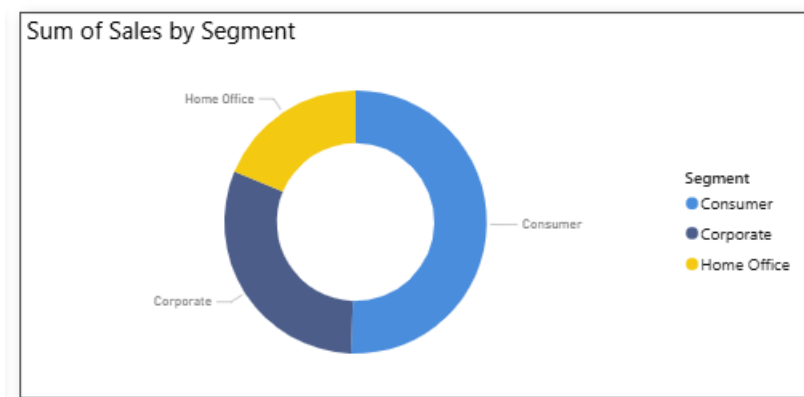
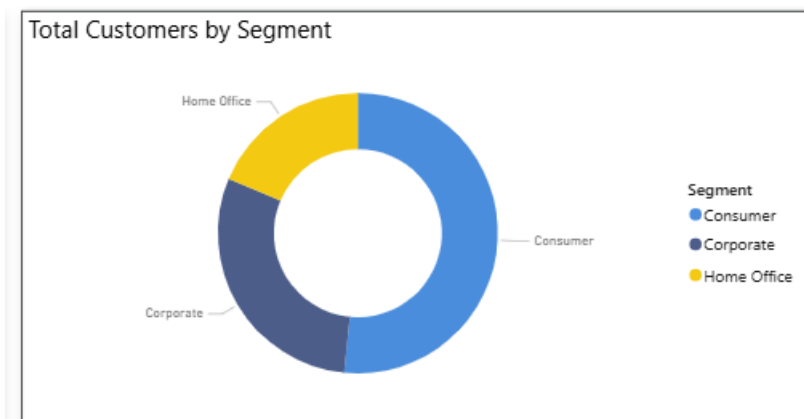
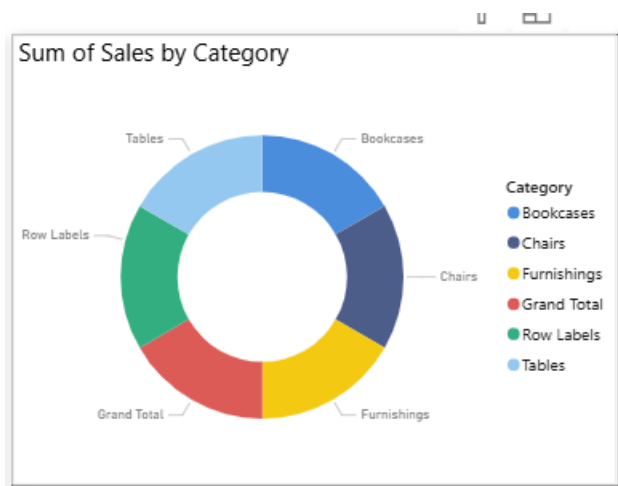
Donut Charts are used to show percentage distribution and category-wise contribution of sales, profit, customers, and products.

The charts were used to analyze:

- Sales Contribution by Region
- Profit Contribution by Region
- Product Sales by Category
- Customer Revenue by Segment
- Quantity Distribution by Category

These visuals help identify dominant product categories, customer segments, regions, and business contributors while providing a clear view of overall distribution patterns.

SCREENSHOT



MAP VISUALIZATION

Purpose:

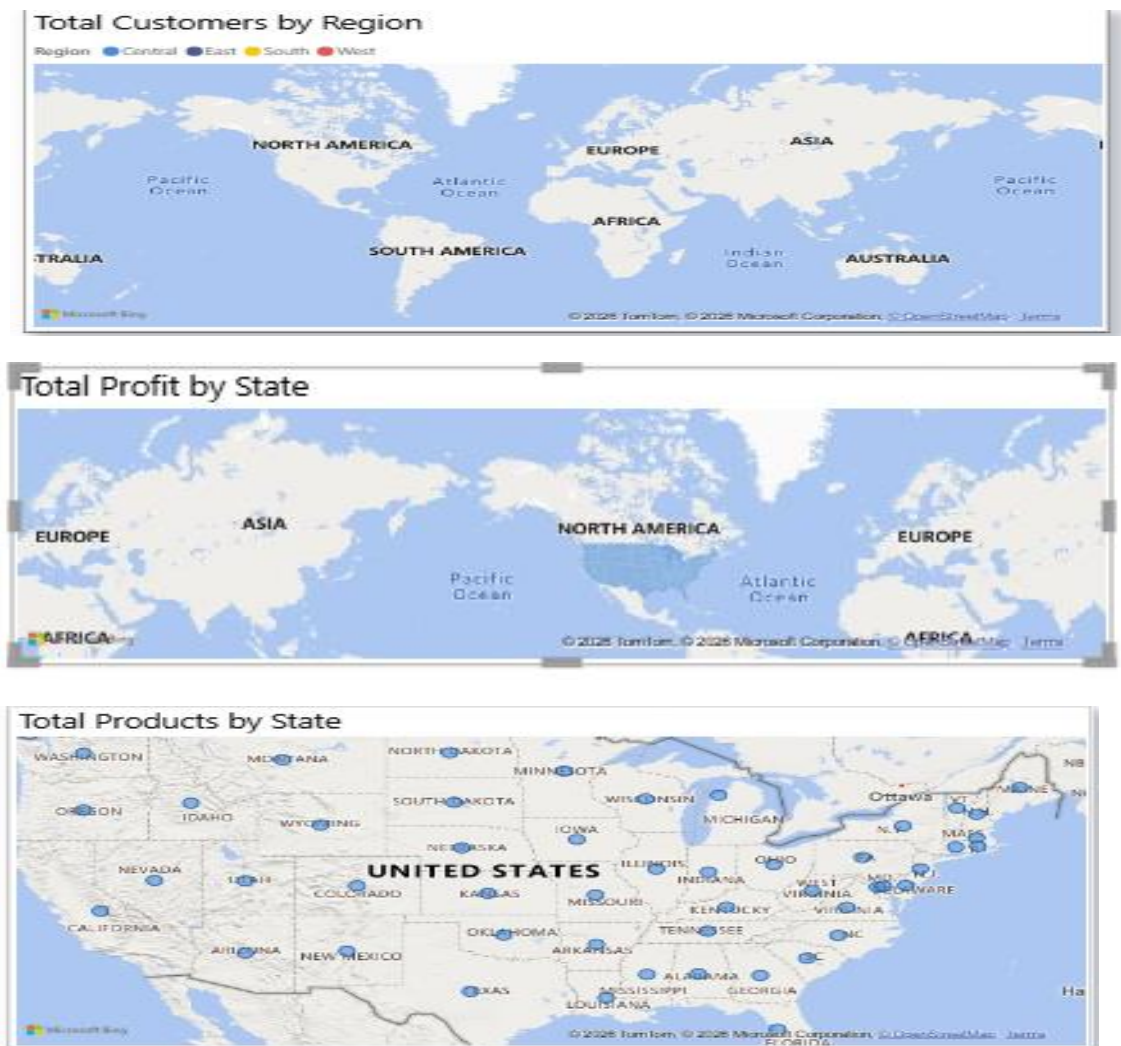
Map Visualizations are used to display the geographical distribution of sales, profit, customers, and business performance across different locations.

The charts were used to analyze:

- Sales Distribution Across States
- Profit Distribution Across States
- Customer Distribution by State
- Regional Sales Performance

These visuals provide location-based insights, help identify high-performing and low-performing regions, and support geographical business analysis and decision-making.

SCREENSHOT



SLICERS

Purpose:

Slicers provide interactive filtering and dynamic analysis across all dashboards, enabling users to explore sales, profit, customer, and product performance from different perspectives.

Slicers Implemented:

Region Slicer

- Enables region-wise analysis of sales and profit performance.

Category Slicer

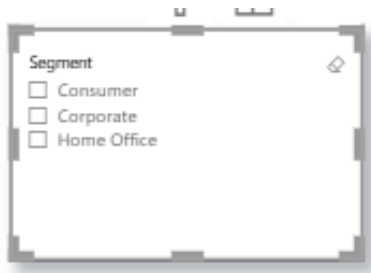
- Filters data by product category.

Segment Slicer

- Enables customer segment-wise analysis.

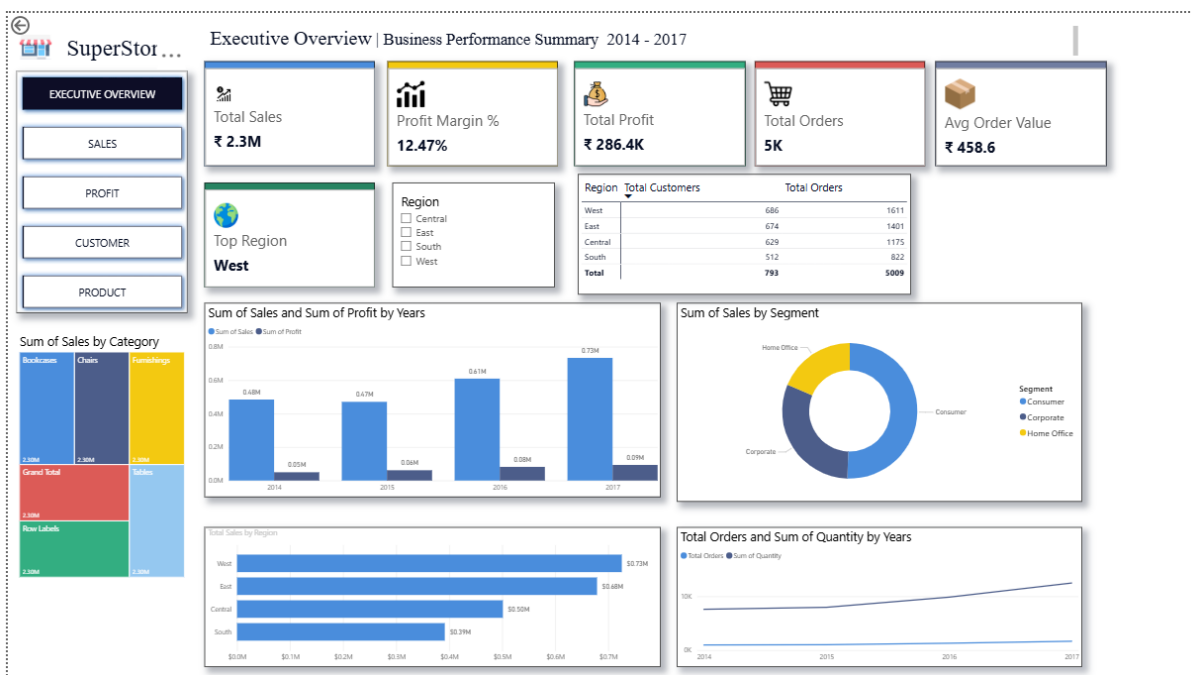
All slicers are synchronized across dashboards using Sync Slicers, ensuring a consistent and interactive user experience throughout the report.

SCREENSHOT



DASHBOARD SCREENSHOTS

EXECUTIVE OVERVIEW



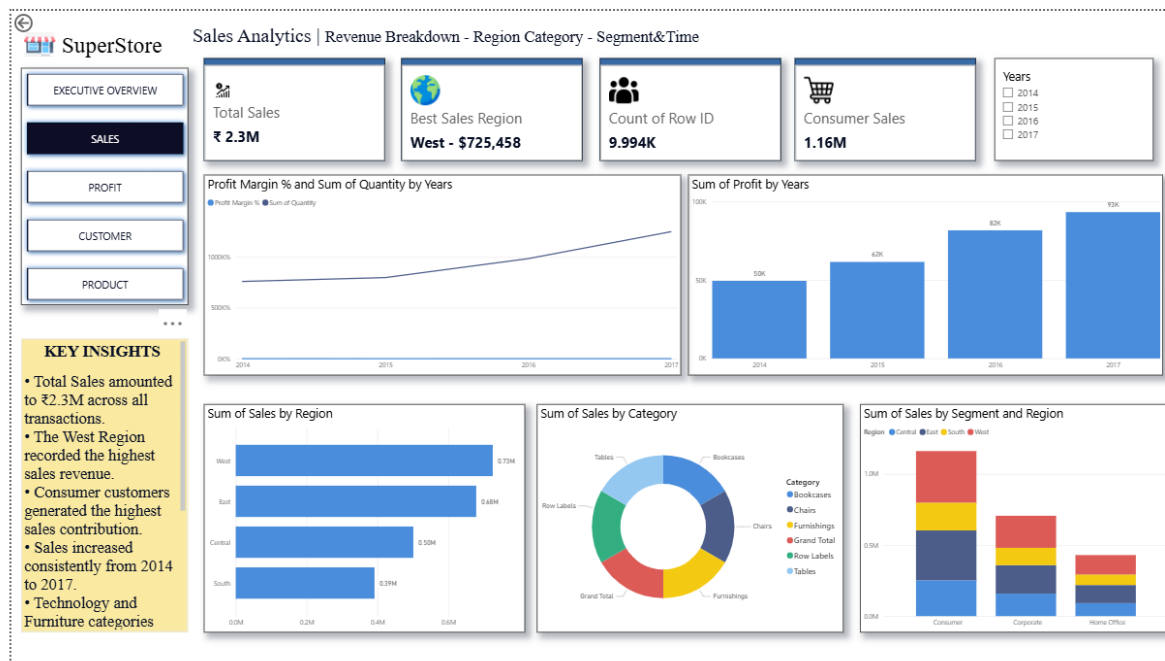
Explanation

The Executive Overview Dashboard provides a high-level summary of overall business performance. It combines key sales, profit, customer, and order metrics to help stakeholders quickly assess the organization's performance and identify major business trends.

Key Insights:

- Total Sales reached ₹2.3M during the analysis period.
- The business generated a Total Profit of ₹286.4K with a Profit Margin of 12.47%.
- The West Region emerged as the top-performing region in terms of sales.
- Sales and profit showed consistent growth from 2014 to 2017.
- Consumer Segment contributed the largest share of total sales.
- Order volume and quantity sold increased steadily over the years.

SALES ANALYTICS



Explanation

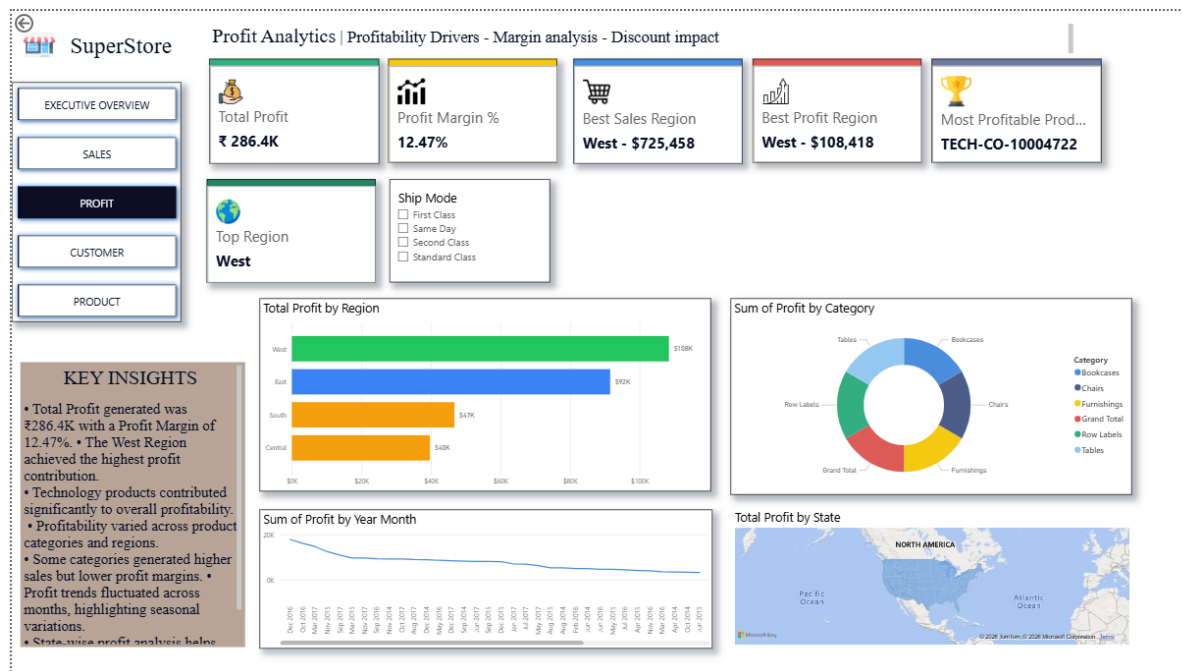
The Sales Analytics Dashboard focuses on analyzing revenue performance across regions, customer segments, product categories, and time periods. It helps identify sales drivers and growth opportunities.

Key Insights:

- Total Sales amounted to ₹2.3M across all transactions.
- The West Region recorded the highest sales revenue.
- Consumer customers generated the highest sales contribution.
- Sales increased consistently from 2014 to 2017.
- Technology and Furniture categories contributed significantly to revenue.
- Regional sales performance varied considerably across different regions.
- Interactive filters enable year-wise and region-wise sales analysis.

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PROFIT ANALYTICS



Explanation

The Profit Analytics Dashboard focuses on profitability analysis across regions, categories, products, and geographical locations. It helps identify profit drivers and areas requiring improvement.

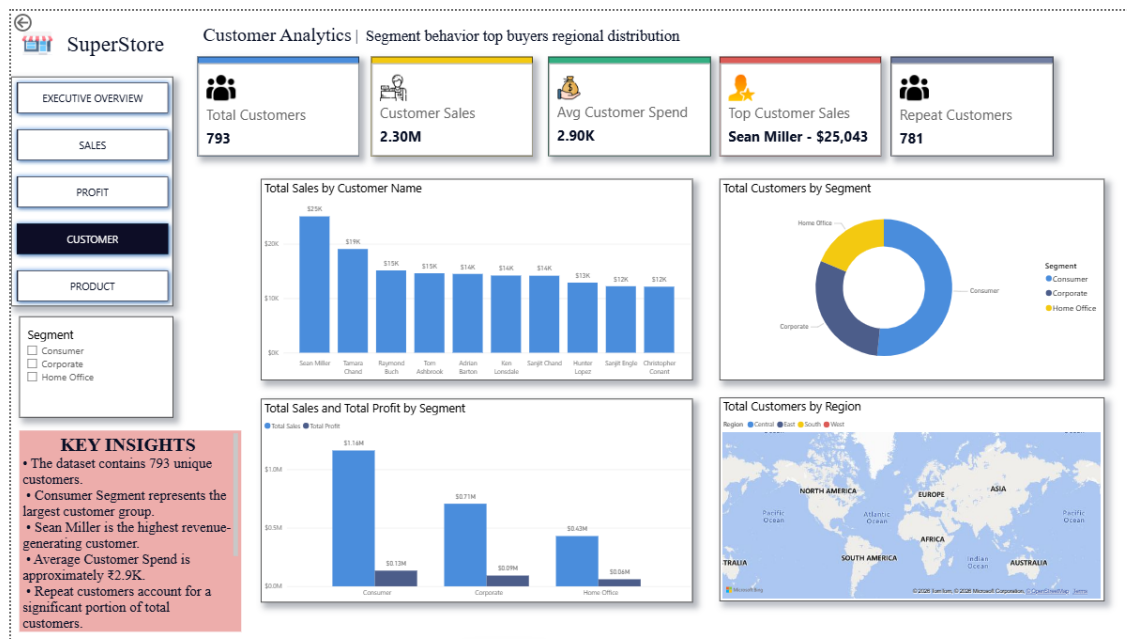
Key Insights:

- Total Profit generated was ₹286.4K with a Profit Margin of 12.47%.
- The West Region achieved the highest profit contribution.
- Technology products contributed significantly to overall profitability.
- Profitability varied across product categories and regions.
- Some categories generated higher sales but lower profit margins.
- Profit trends fluctuated across months, highlighting seasonal variations.
- State-wise profit analysis helps identify profitable and underperforming markets.

CUSTOMER ANALYTICS

Explanation

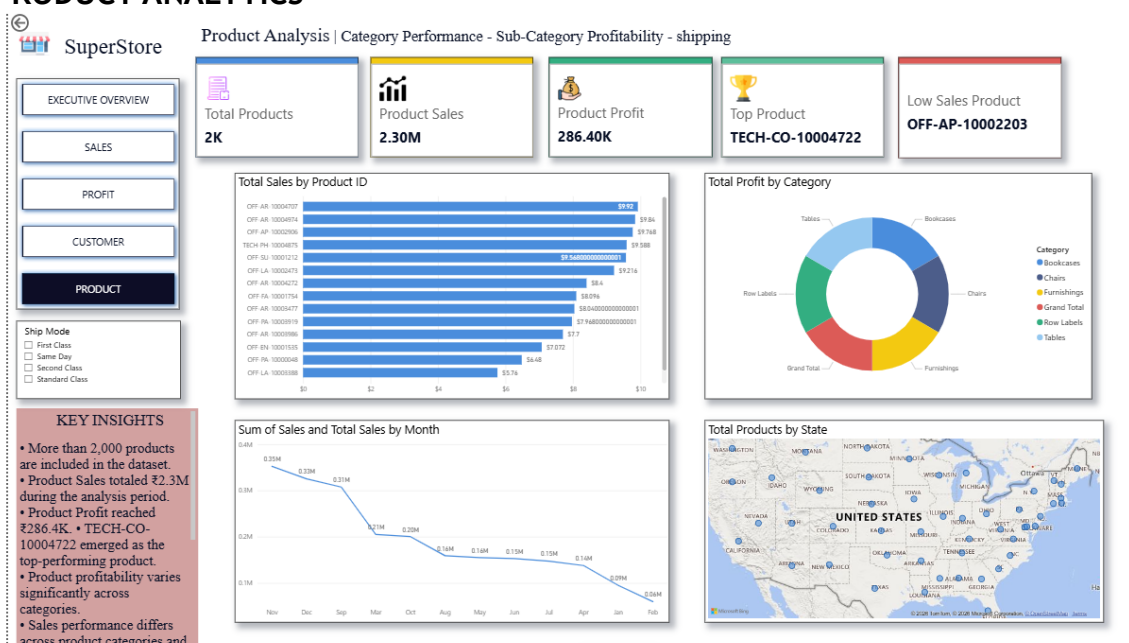
The Customer Analytics Dashboard analyzes customer behavior, spending patterns, segment contribution, and regional customer distribution. It helps understand customer value and purchasing trends.



Key Insights:

- The dataset contains 793 unique customers.
- Consumer Segment represents the largest customer group.
- Sean Miller is the highest revenue-generating customer.
- Average Customer Spend is approximately ₹2.9K.
- Repeat customers account for a significant portion of total customers.
- Customer distribution varies across different regions.
- Consumer customers contribute the majority of overall sales revenue.
- Interactive filtering enables detailed segment-wise customer analysis.

PRODUCT ANALYTICS



Explanation

The Product Analytics Dashboard focuses on product performance, category contribution, profitability, and sales trends. It helps identify top-performing products and profitable categories.

Key Insights:

- More than 2,000 products are included in the dataset.
- Product Sales totaled ₹2.3M during the analysis period.
- Product Profit reached ₹286.4K.
- TECH-CO-10004722 emerged as the top-performing product.
- Product profitability varies significantly across categories.
- Sales performance differs across product categories and states.
- Monthly sales trends highlight changes in product demand over time.
- Interactive filters enable detailed product and shipping mode analysis.

KEY INSIGHTS

INSIGHT-1

The dataset contains approximately 9,994 sales transactions, 793 customers, and 2,000+ products, providing a comprehensive view of sales performance and business operations.

INSIGHT-2

The West Region generates the highest sales and profit, making it the best-performing region in the business.

INSIGHT-3

The Consumer Segment contributes the largest share of total sales, indicating its importance to overall business revenue.

INSIGHT-4

Sales and profit show a consistent upward trend from 2014 to 2017, reflecting business growth over time.

INSIGHT-5

The business achieved a total sales revenue of ₹2.3M and a total profit of ₹286.4K with a profit margin of approximately 12.47%.

INSIGHT-6

Product performance varies significantly across categories and sub-categories, with certain products contributing substantially more revenue and profit.

INSIGHT-7

Profitability is uneven across regions and states, highlighting opportunities for targeted business improvements.

INSIGHT-8

Customer spending patterns differ considerably, with a small group of customers contributing a significant portion of total revenue.

INSIGHT-9

Discounts have a direct impact on profitability, and excessive discounting can reduce profit margins.

INSIGHT-10

State-level analysis reveals geographical differences in sales and profit performance across the market.

INSIGHT-11

Interactive dashboards, slicers, drill-downs, and tooltips enable detailed exploration of business performance and customer behavior.

INSIGHT-12

The integrated dashboard provides actionable insights into sales, profit, customer, product, and regional performance, supporting strategic business decisions.

BUSINESS RECOMMENDATIONS

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RECOMMENDATION-1

Focus marketing and sales efforts on high-performing regions while developing strategies to improve underperforming regions.

RECOMMENDATION-2

Strengthen relationships with high-value customers through loyalty programs and personalized offers.

RECOMMENDATION-3

Optimize discount strategies to maintain profitability while remaining competitive.

RECOMMENDATION-4

Increase inventory and promotional efforts for top-performing products and categories.

RECOMMENDATION-5

Regularly monitor sales trends, customer behavior, and profitability metrics to support data-driven business planning.

FEATURES IMPLEMENTED

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- ✓ KPI Cards
- ✓ Drill Down
- ✓ Drill Through
- ✓ Custom Tooltips

- ✓ Dynamic Titles
- ✓ Slicers
- ✓ Sync Slicers
- ✓ Conditional Formatting
- ✓ Navigation Buttons
- ✓ Interactive Maps
- ✓ Donut Charts
- ✓ Bar Charts
- ✓ Line Charts
- ✓ Cross Filtering
- ✓ Custom Theme

CHALLENGES FACED

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Data Cleaning Challenges
Handling Missing Values
DAX Measure Development
Dynamic Title Implementation
Drill Through Configuration
Tooltip Design
Dashboard Layout Design
Visual Selection and Optimization
Performance Tuning
Data Modeling Challenges

LEARNINGS

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Power Query
Data Cleaning
Data Modeling
DAX Development
Dashboard Design
Interactive Reporting
Storytelling with Data
Business Analysis
Power BI Service Concepts

PROJECT STORYTELLING

BUSINESS SCENARIO

Retail businesses generate large volumes of sales data involving customers, products, orders, profits, and regional operations. However, extracting meaningful insights from raw transactional data is often challenging.

This project transforms sales data into interactive dashboards that support monitoring, performance analysis, and data-driven decision-making across business functions.

BUSINESS PROBLEM

- Which regions generate the highest and lowest sales?
- Which products contribute the most revenue and profit?
- Which customers generate the highest sales?
- How does profitability vary across regions and categories?
- What is the impact of discounts on profit?
- Which regions require business improvement?
- How can sales and profitability be optimized?

ANALYSIS APPROACH

- Analyzed sales performance across regions and states.
- Evaluated customer purchasing behavior and revenue contribution.
- Assessed product sales and profitability.
- Examined category and sub-category performance.
- Analyzed discount impact on profitability.
- Identified top-performing customers and products.
- Implemented Drill Down, Drill Through, Tooltips, Dynamic Titles, and Formatting.
- Developed interactive dashboards for business decision-making.

KEY FINDINGS

- The dataset contains approximately 9,994 transactions and 793 customers.
- Total Sales reached ₹2.3M with Total Profit of ₹286.4K.
- The West Region generated the highest sales and profit.
- Consumer Segment contributed the largest share of revenue.
- Sales and profit increased steadily between 2014 and 2017.
- Product performance varies significantly across categories.
- Discount strategies directly affect profitability.

- State-level performance differs considerably across locations.
- Interactive dashboard analysis improves business visibility and strategic planning.

BUSINESS IMPACT

- Supports data-driven business planning.
- Improves sales performance monitoring.
- Enhances profitability analysis.
- Helps identify high-value customers.
- Assists in product performance evaluation.
- Enables efficient resource allocation and strategy development.
- Provides a centralized business analytics platform.

PROJECT DEPLOYMENT

DEPLOYMENT OBJECTIVE

Deploy the Power BI solution to provide stakeholders with access to interactive sales analytics dashboards and business insights.

DEPLOYMENT PLATFORM

Power BI Service

DEPLOYMENT PROCESS

STEP-1

Validate Data Model and DAX Measures in Power BI Desktop.

STEP-2

Optimize Dashboard Layout and Performance.

STEP-3

Publish PBIX File to Power BI Service.

STEP-4

Configure Dataset Refresh Settings.

STEP-5

Create Dashboards from Published Reports.

STEP-6

Configure User Access and Security Settings.

STEP-7

Share Dashboards with Stakeholders.

BUSINESS BENEFITS

- Centralized Business Analytics Platform
- Interactive Dashboard Reporting
- Improved Sales Monitoring
- Better Profitability Analysis
- Enhanced Customer Insights
- Improved Product Performance Tracking
- Faster Data-Driven Decision Making

CONCLUSION

The **Superstore Sales Analytics Dashboard** successfully transformed raw sales data into interactive and insightful visualizations. The project analyzed sales performance, profitability, customer behavior, product performance, and regional business trends across multiple dimensions.

Through the use of Power BI features such as KPI Cards, Drill Down, Drill Through, Tooltips, Dynamic Titles, Conditional Formatting, Interactive Maps, and Dashboard Navigation, the solution provides valuable insights for business managers and decision-makers.

The dashboard supports informed decision-making, improved profitability, enhanced customer understanding, and strategic business growth through comprehensive data analysis and reporting.